

```
mysql>
mysql>
mysql> /* Numeric functions
/*> abs (): absolute value of number.
/*> ceil(): round up
/*> floor(): round down
/*> round(): round to a specific number of digits.
/*> mod(): modulo of a number
/*> pow()/power(): exponentiation.
/*> sqrt():square root.
/*> exp():exponential function.
/*> radians():convert degrees to radians.
/*> degrees():convert radians to degrees.
/*> sign(): returns the sign of a number.
/*> rand(): generate a random number.
/*> greatest(): find the maximum value.
/*> least(): find the minimum value.
/*> pi(): return the value of p
/*> hex():convert to hexadecimal.
/*> oct(): convert to octal
/*> bin(): convert to binary
/*> */
```

```
mysql> select abs(-100) as absolute_value;
```

```
+-----+
| absolute_value |
+-----+
|          100 |
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> select ceil(13.56) as round_up;
```

```
+-----+
| round_up |
+-----+
|        14 |
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> select floor(12.34) as round_down;
```

```
+-----+
| round_down |
+-----+
|         12 |
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> select round(12.3456,3) rounded_value;
```

```
+-----+
| rounded_value |
```

```
+-----+
|      12.346      |
+-----+
```

1 row in set (0.00 sec)

```
mysql> select mod(13,7) as reminder_value;
```

```
+-----+
| reminder_value |
+-----+
|              6 |
+-----+
```

1 row in set (0.00 sec)

```
mysql> select pow(2,6) as power_value;
```

```
+-----+
| power_value |
+-----+
|          64 |
+-----+
```

1 row in set (0.00 sec)

```
mysql> select sqrt(25) as square_root_value;
```

```
+-----+
| square_root_value |
+-----+
|                  5 |
+-----+
```

1 row in set (0.00 sec)

```
mysql> select ln(8) as natural_log;
```

```
+-----+
| natural_log      |
+-----+
| 2.0794415416798357 |
+-----+
```

1 row in set (0.00 sec)

```
mysql> select log(10,100) as log_value;
```

```
+-----+
| log_value |
+-----+
|          2 |
+-----+
```

1 row in set (0.00 sec)

```
mysql> select log(11,121) as log_value;
```

```
+-----+
| log_value |
+-----+
|          2 |
+-----+
```

```
+-----+
1 row in set (0.00 sec)
```

```
mysql> select log(6,30) as log_value;
```

```
+-----+
| log_value          |
+-----+
| 1.8982444017039273 |
+-----+
1 row in set (0.00 sec)
```

```
mysql> select pi() as pi_value;
```

```
+-----+
| pi_value |
+-----+
| 3.141593 |
+-----+
1 row in set (0.00 sec)
```

```
mysql> select radians(240) as radians_value;
```

```
+-----+
| radians_value      |
+-----+
| 4.1887902047863905 |
+-----+
1 row in set (0.00 sec)
```

```
mysql> select degrees(240) as degrees_value;
```

```
+-----+
| degrees_value      |
+-----+
| 13750.987083139757 |
+-----+
1 row in set (0.00 sec)
```

```
mysql> select degrees(pi())as degrees_value;
```

```
+-----+
| degrees_value      |
+-----+
|          180      |
+-----+
1 row in set (0.00 sec)
```

```
mysql> select sign(-340) as sign_value;
```

```
+-----+
| sign_value |
+-----+
|          -1 |
+-----+
1 row in set (0.00 sec)
```

```
mysql> select sign(340) as sign_value;
```

```
+-----+  
| sign_value |  
+-----+  
|          1 |  
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> select sign(0) as sign_value;
```

```
+-----+  
| sign_value |  
+-----+  
|          0 |  
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> select rand() as random_number;
```

```
+-----+  
| random_number |  
+-----+  
| 0.9954504538285085 |  
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> select rand() as random_number;
```

```
+-----+  
| random_number |  
+-----+  
| 0.6890671017477914 |  
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> select rand() as random_number;
```

```
+-----+  
| random_number |  
+-----+  
| 0.4589841779870765 |  
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> select rand() as random_number;
```

```
+-----+  
| random_number |  
+-----+  
| 0.2277196154256534 |  
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> select greatest(10,20,30,40,50,70) as greatest_value;
```

```

+-----+
| greatest_value |
+-----+
|          70 |
+-----+

```

1 row in set (0.00 sec)

```
mysql> select least(5,10,20,30,40,50,70) as least_value;
```

```

+-----+
| least_value |
+-----+
|          5 |
+-----+

```

1 row in set (0.00 sec)

```
mysql> select bit_count(10) as bit_count;
```

```

+-----+
| bit_count |
+-----+
|          2 |
+-----+

```

1 row in set (0.00 sec)

```
mysql> select bit_count(2) as bit_count;
```

```

+-----+
| bit_count |
+-----+
|          1 |
+-----+

```

1 row in set (0.00 sec)

```
mysql> select bit_count(1) as bit_count;
```

```

+-----+
| bit_count |
+-----+
|          1 |
+-----+

```

1 row in set (0.00 sec)

```
mysql> select bit_count(99) as bit_count;
```

```

+-----+
| bit_count |
+-----+
|          4 |
+-----+

```

1 row in set (0.00 sec)

```
mysql> select bin(10) as binary_value;
```

```

+-----+
| binary_value |
+-----+

```

```
+-----+
| 1010  |
+-----+
```

1 row in set (0.03 sec)

mysql> select bin(1) as binary_value;

```
+-----+
| binary_value |
+-----+
| 1           |
+-----+
```

1 row in set (0.00 sec)

mysql> select bin(2) as binary_value;

```
+-----+
| binary_value |
+-----+
| 10          |
+-----+
```

1 row in set (0.00 sec)

mysql> select bin(9) as binary_value;

```
+-----+
| binary_value |
+-----+
| 1001        |
+-----+
```

1 row in set (0.00 sec)

mysql> select oct(64) as octal_value;

```
+-----+
| octal_value |
+-----+
| 100        |
+-----+
```

1 row in set (0.00 sec)

mysql> select oct(56) as octal_value;

```
+-----+
| octal_value |
+-----+
| 70         |
+-----+
```

1 row in set (0.00 sec)

mysql> select hex(255) as hex_value;

```
+-----+
| hex_value |
+-----+
| FF       |
+-----+
```

```
+-----+
1 row in set (0.00 sec)
```

```
mysql> select hex(200) as hex_value;
```

```
+-----+
| hex_value |
+-----+
| C8        |
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> select hex(222) as hex_value;
```

```
+-----+
| hex_value |
+-----+
| DE        |
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> select hex(221) as hex_value;
```

```
+-----+
| hex_value |
+-----+
| DD        |
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> select hex(220) as hex_value;
```

```
+-----+
| hex_value |
+-----+
| DC        |
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> drop table employees;
```

```
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> CREATE TABLE employees (
-> emp_id INT PRIMARY KEY,
-> first_name VARCHAR(30),
-> last_name VARCHAR(30),
-> email VARCHAR(60),
-> phone_number VARCHAR(20),
-> department VARCHAR(30),
-> city VARCHAR(30),
-> salary DECIMAL(10,2)
-> );
```

```
Query OK, 0 rows affected (0.06 sec)
```

```

mysql>
mysql>
mysql> INSERT INTO employees VALUES
    ->
(101,'Rahul','Kumar','rahul.kumar@gmail.com','987-654-3210','IT','Hyderabad',45000)
,
    ->
(102,'Anita','Sharma','anita.sharma@yahoo.com','912-345-6789','HR','Bangalore',38000),
    ->
(103,'Suresh','Verma','suresh.verma@gmail.com','998-123-4567','Sales','Chennai',52000),
    ->
(104,'Priya','Singh','priya.singh@outlook.com','900-456-7890','Finance','Pune',47000),
    ->
(105,'Rahul','Joshi','rahul.joshi@gmail.com','955-888-1111','Testing','Delhi',41000),
    ->
(106,'Kiran','Reddy','kiran.reddy@gmail.com','901-222-3333','IT','Hyderabad',56000)
,
    ->
(107,'Deepika','Naidu','deepika.naidu@yahoo.com','944-111-2222','HR','Vizag',39000)
,
    ->
(108,'Mahesh','Patel','mahesh.patel@gmail.com','933-444-5555','Sales','Mumbai',62000),
    ->
(109,'Divya','Gupta','divya.gupta@gmail.com','955-777-8888','Finance','Delhi',49000)
),
    ->
(110,'Arjun','Rao','arjun.rao@gmail.com','988-999-7777','Developer','Hyderabad',68000);
Query OK, 10 rows affected (0.02 sec)
Records: 10  Duplicates: 0  Warnings: 0

```

```
mysql> select * from employees;
```

emp_id	first_name	last_name	email	phone_number	department	city	salary
101	Rahul	Kumar	rahul.kumar@gmail.com	987-654-3210	IT	Hyderabad	45000.00
102	Anita	Sharma	anita.sharma@yahoo.com	912-345-6789	HR	Bangalore	38000.00
103	Suresh	Verma	suresh.verma@gmail.com	998-123-4567	Sales	Chennai	52000.00
104	Priya	Singh	priya.singh@outlook.com	900-456-7890			


```
+-----+
10 rows in set (0.00 sec)
```

```
mysql> -- lower funtion
```

```
mysql> select last_name, lower(last_name)as lower_name from employees;
```

```
+-----+-----+
| last_name | lower_name |
+-----+-----+
| Kumar     | kumar     |
| Sharma    | sharma    |
| Verma     | verma     |
| Singh     | singh     |
| Joshi     | joshi     |
| Reddy     | reddy     |
| Naidu     | naidu     |
| Patel     | patel     |
| Gupta     | gupta     |
| Rao       | rao       |
+-----+-----+
```

```
10 rows in set (0.00 sec)
```

```
mysql> select email,substring(email,1,10) as email_prefix from employees;
```

```
+-----+-----+
| email                | email_prefix |
+-----+-----+
| rahul.kumar@gmail.com | rahul.kuma   |
| anita.sharma@yahoo.com | anita.shar   |
| suresh.verma@gmail.com | suresh.ver   |
| priya.singh@outlook.com | priya.sing   |
| rahul.joshi@gmail.com  | rahul.josh   |
| kiran.reddy@gmail.com  | kiran.redd   |
| deepika.naidu@yahoo.com | deepika.na   |
| mahesh.patel@gmail.com  | mahesh.pat   |
| divya.gupta@gmail.com   | divya.gupt   |
| arjun.rao@gmail.com     | arjun.rao@   |
+-----+-----+
```

```
10 rows in set (0.00 sec)
```

```
mysql> select first_name,length(first_name)as first_name_length from employees;
```

```
+-----+-----+
| first_name | first_name_length |
+-----+-----+
| Rahul     | 5 |
| Anita     | 5 |
| Suresh    | 6 |
| Priya     | 5 |
| Rahul     | 5 |
| Kiran     | 5 |
| Deepika   | 7 |
| Mahesh    | 6 |
+-----+-----+
```

Divya	5
Arjun	5

10 rows in set (0.00 sec)

```
mysql> select city,length(city)as length_of_city from employees;
```

city	length_of_city
Hyderabad	9
Bangalore	9
Chennai	7
Pune	4
Delhi	5
Hyderabad	9
Vizag	5
Mumbai	6
Delhi	5
Hyderabad	9

10 rows in set (0.00 sec)

```
mysql> select * from employees;
```

emp_id	first_name	last_name	email	phone_number	department	city	salary
101	Rahul	Kumar	rahul.kumar@gmail.com	987-654-3210	IT	Hyderabad	45000.00
102	Anita	Sharma	anita.sharma@yahoo.com	912-345-6789	HR	Bangalore	38000.00
103	Suresh	Verma	suresh.verma@gmail.com	998-123-4567	Sales	Chennai	52000.00
104	Priya	Singh	priya.singh@outlook.com	900-456-7890	Finance	Pune	47000.00
105	Rahul	Joshi	rahul.joshi@gmail.com	955-888-1111	Testing	Delhi	41000.00
106	Kiran	Reddy	kiran.reddy@gmail.com	901-222-3333	IT	Hyderabad	56000.00
107	Deepika	Naidu	deepika.naidu@yahoo.com	944-111-2222	HR	Vizag	39000.00
108	Mahesh	Patel	mahesh.patel@gmail.com	933-444-5555	Sales	Mumbai	62000.00
109	Divya	Gupta	divya.gupta@gmail.com	955-777-8888	Finance	Delhi	49000.00
110	Arjun	Rao	arjun.rao@gmail.com	988-999-7777	Developer	Hyderabad	68000.00

```

+-----+-----+-----+-----+-----+-----+-----+
-----+-----+-----+
10 rows in set (0.00 sec)

```

```

mysql> -- replace
mysql> select phone_number,replace(phone_number,'-','')as clean_phone from
employees;

```

```

+-----+-----+
| phone_number | clean_phone |
+-----+-----+
| 987-654-3210 | 9876543210 |
| 912-345-6789 | 9123456789 |
| 998-123-4567 | 9981234567 |
| 900-456-7890 | 9004567890 |
| 955-888-1111 | 9558881111 |
| 901-222-3333 | 9012223333 |
| 944-111-2222 | 9441112222 |
| 933-444-5555 | 9334445555 |
| 955-777-8888 | 9557778888 |
| 988-999-7777 | 9889997777 |
+-----+-----+
10 rows in set (0.00 sec)

```

```

mysql> insert into employees(emp_id,first_name) values(111,' Arun Kumar ');
Query OK, 1 row affected (0.01 sec)

```

```

mysql> select * from employees;

```

```

+-----+-----+-----+-----+-----+-----+-----+
-----+-----+-----+
| emp_id | first_name | last_name | email | phone_number |
department | city | salary |
+-----+-----+-----+-----+-----+-----+-----+
| 101 | Rahul | Kumar | rahul.kumar@gmail.com | 987-654-3210 | IT
| | Hyderabad | 45000.00 |
| 102 | Anita | Sharma | anita.sharma@yahoo.com | 912-345-6789 | HR
| | Bangalore | 38000.00 |
| 103 | Suresh | Verma | suresh.verma@gmail.com | 998-123-4567 |
Sales | Chennai | 52000.00 |
| 104 | Priya | Singh | priya.singh@outlook.com | 900-456-7890 |
Finance | Pune | 47000.00 |
| 105 | Rahul | Joshi | rahul.joshi@gmail.com | 955-888-1111 |
Testing | Delhi | 41000.00 |
| 106 | Kiran | Reddy | kiran.reddy@gmail.com | 901-222-3333 | IT
| | Hyderabad | 56000.00 |
| 107 | Deepika | Naidu | deepika.naidu@yahoo.com | 944-111-2222 | HR
| | Vizag | 39000.00 |
| 108 | Mahesh | Patel | mahesh.patel@gmail.com | 933-444-5555 |
Sales | Mumbai | 62000.00 |
| 109 | Divya | Gupta | divya.gupta@gmail.com | 955-777-8888 |

```

Finance	Delhi	49000.00			
110	Arjun	Rao	arjun.rao@gmail.com	988-999-7777	
Developer	Hyderabad	68000.00			
111	Arun Kumar	NULL	NULL	NULL	NULL
	NULL	NULL			

```

+-----+-----+-----+-----+-----+-----+
-----+-----+-----+
11 rows in set (0.00 sec)

```

```
mysql> select first_name,trim(first_name) as trimmed_name from employees;
```

first_name	trimmed_name
Rahul	Rahul
Anita	Anita
Suresh	Suresh
Priya	Priya
Rahul	Rahul
Kiran	Kiran
Deepika	Deepika
Mahesh	Mahesh
Divya	Divya
Arjun	Arjun
Arun Kumar	Arun Kumar

```

+-----+-----+
11 rows in set (0.00 sec)

```

```
mysql> select concat_ws(',',first_name,last_name,department) as emp_info from employees;
```

emp_info
Rahul,Kumar,IT
Anita,Sharma,HR
Suresh,Verma,Sales
Priya,Singh,Finance
Rahul,Joshi,Testing
Kiran,Reddy,IT
Deepika,Naidu,HR
Mahesh,Patel,Sales
Divya,Gupta,Finance
Arjun,Rao,Developer
Arun Kumar

```

+-----+
11 rows in set (0.00 sec)

```

```
mysql> select first_name,position('a' in first_name)as postion_of_A from employees;
```

first_name	postion_of_A
------------	--------------

Rahul	2
Anita	1
Suresh	0
Priya	5
Rahul	2
Kiran	4
Deepika	7
Mahesh	2
Divya	5
Arjun	1
Arun Kumar	2

+-----+

11 rows in set (0.00 sec)

mysql> select last_name,position('a' in last_name)as postion_of_A from employees;

last_name	postion_of_A
Kumar	4
Sharma	3
Verma	5
Singh	0
Joshi	0
Reddy	0
Naidu	2
Patel	2
Gupta	5
Rao	2
NULL	NULL

+-----+

11 rows in set (0.00 sec)

mysql> select first_name,left(first_name,3)as short_name from employees;

first_name	short_name
Rahul	Rah
Anita	Ani
Suresh	Sur
Priya	Pri
Rahul	Rah
Kiran	Kir
Deepika	Dee
Mahesh	Mah
Divya	Div
Arjun	Arj
Arun Kumar	Ar

+-----+

11 rows in set (0.00 sec)

```
mysql> select first_name,right(first_name,3)as ended_name from employees;
```

first_name	ended_name
Rahul	hul
Anita	ita
Suresh	esh
Priya	iya
Rahul	hul
Kiran	ran
Deepika	ika
Mahesh	esh
Divya	vya
Arjun	jun
Arun Kumar	ar

```
11 rows in set (0.00 sec)
```

```
mysql> select last_name,reverse(last_name)as revested_name from employees;
```

last_name	revested_name
Kumar	ramuK
Sharma	amrahS
Verma	amreV
Singh	hgniS
Joshi	ihsoJ
Reddy	yddeR
Naidu	udiaN
Patel	letaP
Gupta	atpuG
Rao	oaR
NULL	NULL

```
11 rows in set (0.00 sec)
```

```
mysql> select first_name,reverse(first_name)as revested_name from employees;
```

first_name	revested_name
Rahul	luhaR
Anita	atinA
Suresh	hseruS
Priya	ayirP
Rahul	luhaR
Kiran	nariK
Deepika	akipeeD
Mahesh	hsehaM
Divya	ayviD
Arjun	nujrA

Arun Kumar	ramuK nurA
------------	------------

11 rows in set (0.00 sec)

```
mysql> select first_name,lpad(first_name,10,'*')as padded_name from employees;
```

first_name	padded_name
Rahul	*****Rahul
Anita	*****Anita
Suresh	****Suresh
Priya	*****Priya
Rahul	*****Rahul
Kiran	*****Kiran
Deepika	***Deepika
Mahesh	****Mahesh
Divya	*****Divya
Arjun	*****Arjun
Arun Kumar	Arun Kuma

11 rows in set (0.00 sec)

```
mysql> select first_name,rpad(first_name,10,'*')as padded_name from employees;
```

first_name	padded_name
Rahul	Rahul*****
Anita	Anita*****
Suresh	Suresh*****
Priya	Priya*****
Rahul	Rahul*****
Kiran	Kiran*****
Deepika	Deepika***
Mahesh	Mahesh*****
Divya	Divya*****
Arjun	Arjun*****
Arun Kumar	Arun Kuma

11 rows in set (0.00 sec)

```
mysql> select first_name,rpad(first_name,10,'?')as padded_name from employees;
```

first_name	padded_name
Rahul	Rahul?????
Anita	Anita?????
Suresh	Suresh?????
Priya	Priya?????
Rahul	Rahul?????
Kiran	Kiran?????

Deepika	Deepika???
Mahesh	Mahesh????
Divya	Divya?????
Arjun	Arjun?????
Arun Kumar	Arun Kuma

11 rows in set (0.00 sec)

```
mysql> select first_name,ascii(first_name)as ascii_value from employees;
```

first_name	ascii_value
Rahul	82
Anita	65
Suresh	83
Priya	80
Rahul	82
Kiran	75
Deepika	68
Mahesh	77
Divya	68
Arjun	65
Arun Kumar	32

11 rows in set (0.00 sec)

```
mysql> select city,ascii(city)as ascii_value from employees;
```

city	ascii_value
Hyderabad	72
Bangalore	66
Chennai	67
Pune	80
Delhi	68
Hyderabad	72
Vizag	86
Mumbai	77
Delhi	68
Hyderabad	72
NULL	NULL

11 rows in set (0.00 sec)

```
mysql> select * from employees;
```

emp_id	first_name	last_name	email	phone_number
department	city	salary		

101	Rahul	Kumar	rahul.kumar@gmail.com	987-654-3210	IT
	Hyderabad	45000.00			
102	Anita	Sharma	anita.sharma@yahoo.com	912-345-6789	HR
	Bangalore	38000.00			
103	Suresh	Verma	suresh.verma@gmail.com	998-123-4567	
Sales	Chennai	52000.00			
104	Priya	Singh	priya.singh@outlook.com	900-456-7890	
Finance	Pune	47000.00			
105	Rahul	Joshi	rahul.joshi@gmail.com	955-888-1111	
Testing	Delhi	41000.00			
106	Kiran	Reddy	kiran.reddy@gmail.com	901-222-3333	IT
	Hyderabad	56000.00			
107	Deepika	Naidu	deepika.naidu@yahoo.com	944-111-2222	HR
	Vizag	39000.00			
108	Mahesh	Patel	mahesh.patel@gmail.com	933-444-5555	
Sales	Mumbai	62000.00			
109	Divya	Gupta	divya.gupta@gmail.com	955-777-8888	
Finance	Delhi	49000.00			
110	Arjun	Rao	arjun.rao@gmail.com	988-999-7777	
Developer	Hyderabad	68000.00			
111	Arun Kumar	NULL	NULL	NULL	NULL
	NULL	NULL			

+-----+-----+-----+-----+-----+-----+-----+

-----+-----+-----+-----+

11 rows in set (0.00 sec)

```
mysql> select first_name from employees where first_name like "r%";
```

first_name
Rahul
Rahul

2 rows in set (0.00 sec)

```
mysql> select email from employees where email like '%@yahoo.com';
```

email
anita.sharma@yahoo.com
deepika.naidu@yahoo.com

2 rows in set (0.00 sec)

```
mysql> select email,length(email)as length_of_email from employees;
```

email	length_of_email
-------	-----------------

rahul.kumar@gmail.com	21
anita.sharma@yahoo.com	22
suresh.verma@gmail.com	22
priya.singh@outlook.com	23
rahul.joshi@gmail.com	21
kiran.reddy@gmail.com	21
deepika.naidu@yahoo.com	23
mahesh.patel@gmail.com	22
divya.gupta@gmail.com	21
arjun.rao@gmail.com	19
NULL	NULL

11 rows in set (0.00 sec)

```
mysql> select concat(first_name, ' ', last_name, department) as details from employees;
```

details
Rahul KumarIT
Anita SharmaHR
Suresh VermaSales
Priya SinghFinance
Rahul JoshiTesting
Kiran ReddyIT
Deepika NaiduHR
Mahesh PatelSales
Divya GuptaFinance
Arjun RaoDeveloper
NULL

11 rows in set (0.00 sec)

```
mysql> select first_name, length(first_name)>5 from employees;
```

first_name	length(first_name)>5
Rahul	0
Anita	0
Suresh	1
Priya	0
Rahul	0
Kiran	0
Deepika	1
Mahesh	1
Divya	0
Arjun	0

Arun Kumar	1
------------	---

11 rows in set (0.00 sec)

```
mysql> select email,replace(email,'gmail.com','company.com') from employees;
```

email	replace(email,'gmail.com','company.com')
rahul.kumar@gmail.com	rahul.kumar@company.com
anita.sharma@yahoo.com	anita.sharma@yahoo.com
suresh.verma@gmail.com	suresh.verma@company.com
priya.singh@outlook.com	priya.singh@outlook.com
rahul.joshi@gmail.com	rahul.joshi@company.com
kiran.reddy@gmail.com	kiran.reddy@company.com
deepika.naidu@yahoo.com	deepika.naidu@yahoo.com
mahesh.patel@gmail.com	mahesh.patel@company.com
divya.gupta@gmail.com	divya.gupta@company.com
arjun.rao@gmail.com	arjun.rao@company.com
NULL	NULL

11 rows in set (0.00 sec)

```
mysql> select last_name from employees where last_name like '%a';
```

last_name
Sharma
Verma
Gupta

3 rows in set (0.00 sec)

```
mysql> select email,position('@' in email) from employees;
```

email	position('@' in email)
rahul.kumar@gmail.com	12
anita.sharma@yahoo.com	13
suresh.verma@gmail.com	13
priya.singh@outlook.com	12
rahul.joshi@gmail.com	12
kiran.reddy@gmail.com	12
deepika.naidu@yahoo.com	14
mahesh.patel@gmail.com	13
divya.gupta@gmail.com	12
arjun.rao@gmail.com	10
NULL	NULL

11 rows in set (0.00 sec)

```
mysql> select email,substring(email,1,11) as email_prefix from employees;
```

email	email_prefix
rahul.kumar@gmail.com	rahul.kumar
anita.sharma@yahoo.com	anita.sharm
suresh.verma@gmail.com	suresh.verm
priya.singh@outlook.com	priya.singh
rahul.joshi@gmail.com	rahul.joshi
kiran.reddy@gmail.com	kiran.reddy
deepika.naidu@yahoo.com	deepika.nai
mahesh.patel@gmail.com	mahesh.pate
divya.gupta@gmail.com	divya.gupta
arjun.rao@gmail.com	arjun.rao@g
NULL	NULL

```
11 rows in set (0.00 sec)
```

```
mysql> select email,substring(email,1,12) as email_prefix from employees;
```

email	email_prefix
rahul.kumar@gmail.com	rahul.kumar@
anita.sharma@yahoo.com	anita.sharma
suresh.verma@gmail.com	suresh.verma
priya.singh@outlook.com	priya.singh@
rahul.joshi@gmail.com	rahul.joshi@
kiran.reddy@gmail.com	kiran.reddy@
deepika.naidu@yahoo.com	deepika.naid
mahesh.patel@gmail.com	mahesh.patel
divya.gupta@gmail.com	divya.gupta@
arjun.rao@gmail.com	arjun.rao@gm
NULL	NULL

```
11 rows in set (0.00 sec)
```

```
mysql> select email,substring(email,1,10) as email_prefix from employees;
```

email	email_prefix
rahul.kumar@gmail.com	rahul.kuma
anita.sharma@yahoo.com	anita.shar
suresh.verma@gmail.com	suresh.ver
priya.singh@outlook.com	priya.sing
rahul.joshi@gmail.com	rahul.josh
kiran.reddy@gmail.com	kiran.redd
deepika.naidu@yahoo.com	deepika.na
mahesh.patel@gmail.com	mahesh.pat
divya.gupta@gmail.com	divya.gupt

arjun.rao@gmail.com	arjun.rao@
NULL	NULL

11 rows in set (0.00 sec)

```
mysql> select email,substring(email,2,10) as email_prefix from employees;
```

email	email_prefix
rahul.kumar@gmail.com	ahul.kumar
anita.sharma@yahoo.com	nita.sharm
suresh.verma@gmail.com	uresh.verm
priya.singh@outlook.com	riya.singh
rahul.joshi@gmail.com	ahul.joshi
kiran.reddy@gmail.com	iran.reddy
deepika.naidu@yahoo.com	eepika.nai
mahesh.patel@gmail.com	ahesh.pate
divya.gupta@gmail.com	ivya.gupta
arjun.rao@gmail.com	rjun.rao@g
NULL	NULL

11 rows in set (0.00 sec)

```
mysql> select email,substring(email,10)from employees;
```

email	substring(email,10)
rahul.kumar@gmail.com	ar@gmail.com
anita.sharma@yahoo.com	rma@yahoo.com
suresh.verma@gmail.com	rma@gmail.com
priya.singh@outlook.com	gh@outlook.com
rahul.joshi@gmail.com	hi@gmail.com
kiran.reddy@gmail.com	dy@gmail.com
deepika.naidu@yahoo.com	aidu@yahoo.com
mahesh.patel@gmail.com	tel@gmail.com
divya.gupta@gmail.com	ta@gmail.com
arjun.rao@gmail.com	@gmail.com
NULL	NULL

11 rows in set (0.00 sec)

```
mysql> select email,substring(email,12)from employees;
```

email	substring(email,12)
rahul.kumar@gmail.com	@gmail.com
anita.sharma@yahoo.com	a@yahoo.com
suresh.verma@gmail.com	a@gmail.com
priya.singh@outlook.com	@outlook.com
rahul.joshi@gmail.com	@gmail.com

kiran.reddy@gmail.com	@gmail.com
deepika.naidu@yahoo.com	du@yahoo.com
mahesh.patel@gmail.com	l@gmail.com
divya.gupta@gmail.com	@gmail.com
arjun.rao@gmail.com	mail.com
NULL	NULL

11 rows in set (0.00 sec)

mysql> select * from employees;

emp_id	first_name	last_name	email	phone_number	
department	city	salary			
101	Rahul	Kumar	rahul.kumar@gmail.com	987-654-3210	IT
	Hyderabad	45000.00			
102	Anita	Sharma	anita.sharma@yahoo.com	912-345-6789	HR
	Bangalore	38000.00			
103	Suresh	Verma	suresh.verma@gmail.com	998-123-4567	
Sales	Chennai	52000.00			
104	Priya	Singh	priya.singh@outlook.com	900-456-7890	
Finance	Pune	47000.00			
105	Rahul	Joshi	rahul.joshi@gmail.com	955-888-1111	
Testing	Delhi	41000.00			
106	Kiran	Reddy	kiran.reddy@gmail.com	901-222-3333	IT
	Hyderabad	56000.00			
107	Deepika	Naidu	deepika.naidu@yahoo.com	944-111-2222	HR
	Vizag	39000.00			
108	Mahesh	Patel	mahesh.patel@gmail.com	933-444-5555	
Sales	Mumbai	62000.00			
109	Divya	Gupta	divya.gupta@gmail.com	955-777-8888	
Finance	Delhi	49000.00			
110	Arjun	Rao	arjun.rao@gmail.com	988-999-7777	
Developer	Hyderabad	68000.00			
111	Arun Kumar	NULL	NULL	NULL	NULL
	NULL	NULL			

11 rows in set (0.00 sec)

mysql> select phone_number,replace(phone_number,'-','') from employees;

phone_number	replace(phone_number,'-','')
987-654-3210	9876543210
912-345-6789	9123456789
998-123-4567	9981234567
900-456-7890	9004567890

955-888-1111	9558881111
901-222-3333	9012223333
944-111-2222	9441112222
933-444-5555	9334445555
955-777-8888	9557778888
988-999-7777	9889997777
NULL	NULL

11 rows in set (0.00 sec)

```
mysql> select first_name from employees where first_name like 'i%';
Empty set (0.00 sec)
```

```
mysql> select upper(concat(first_name, ' ',last_name)) as upper_full_name from
employees where department='IT';
+-----+
| upper_full_name |
+-----+
| RAHUL KUMAR     |
| KIRAN REDDY     |
+-----+
2 rows in set (0.00 sec)
```

```
mysql> select first_name,count(first_name) as count from employees group by
first_name having count>1;
+-----+-----+
| first_name | count |
+-----+-----+
| Rahul      | 2     |
+-----+-----+
1 row in set (0.01 sec)
```

```
mysql> select * from employees where first_name like '%i%';
+-----+-----+-----+-----+-----+-----+
| emp_id | first_name | last_name | email | phone_number |
+-----+-----+-----+-----+-----+
| 102    | Anita     | Sharma   | anita.sharma@yahoo.com | 912-345-6789 | HR
| 104    | Priya     | Singh    | priya.singh@outlook.com | 900-456-7890 |
Finance | Pune      | 47000.00 |
| 106    | Kiran     | Reddy    | kiran.reddy@gmail.com   | 901-222-3333 | IT
| 107    | Deepika   | Naidu    | deepika.naidu@yahoo.com | 944-111-2222 | HR
| 109    | Divya     | Gupta    | divya.gupta@gmail.com   | 955-777-8888 |
Finance | Delhi     | 49000.00 |
```

```
+-----+-----+-----+-----+-----+-----+-----+
-----+-----+-----+
5 rows in set (0.00 sec)
```

```
mysql> select * from employees order by length(first_name);
```

```
+-----+-----+-----+-----+-----+-----+-----+
-----+-----+-----+
| emp_id | first_name | last_name | email | phone_number |
department | city | salary |
+-----+-----+-----+-----+-----+-----+-----+
-----+-----+-----+
| 101 | Rahul | Kumar | rahul.kumar@gmail.com | 987-654-3210 | IT
| | Hyderabad | 45000.00 |
| 102 | Anita | Sharma | anita.sharma@yahoo.com | 912-345-6789 | HR
| | Bangalore | 38000.00 |
| 104 | Priya | Singh | priya.singh@outlook.com | 900-456-7890 |
Finance | Pune | 47000.00 |
| 105 | Rahul | Joshi | rahul.joshi@gmail.com | 955-888-1111 |
Testing | Delhi | 41000.00 |
| 106 | Kiran | Reddy | kiran.reddy@gmail.com | 901-222-3333 | IT
| | Hyderabad | 56000.00 |
| 109 | Divya | Gupta | divya.gupta@gmail.com | 955-777-8888 |
Finance | Delhi | 49000.00 |
| 110 | Arjun | Rao | arjun.rao@gmail.com | 988-999-7777 |
Developer | Hyderabad | 68000.00 |
| 103 | Suresh | Verma | suresh.verma@gmail.com | 998-123-4567 |
Sales | Chennai | 52000.00 |
| 108 | Mahesh | Patel | mahesh.patel@gmail.com | 933-444-5555 |
Sales | Mumbai | 62000.00 |
| 107 | Deepika | Naidu | deepika.naidu@yahoo.com | 944-111-2222 | HR
| | Vizag | 39000.00 |
| 111 | Arun Kumar | NULL | NULL | NULL | NULL
| | NULL | NULL |
+-----+-----+-----+-----+-----+-----+-----+
-----+-----+-----+
11 rows in set (0.00 sec)
```

```
mysql> select * from employees where length(city)>6;
```

```
+-----+-----+-----+-----+-----+-----+-----+
-----+-----+-----+
| emp_id | first_name | last_name | email | phone_number |
department | city | salary |
+-----+-----+-----+-----+-----+-----+-----+
-----+-----+-----+
| 101 | Rahul | Kumar | rahul.kumar@gmail.com | 987-654-3210 | IT
| | Hyderabad | 45000.00 |
| 102 | Anita | Sharma | anita.sharma@yahoo.com | 912-345-6789 | HR
| | Bangalore | 38000.00 |
| 103 | Suresh | Verma | suresh.verma@gmail.com | 998-123-4567 | Sales
| | Chennai | 52000.00 |
```

	106	Kiran	Reddy	kiran.reddy@gmail.com	901-222-3333	IT
		Hyderabad	56000.00			
	110	Arjun	Rao	arjun.rao@gmail.com	988-999-7777	
Developer		Hyderabad	68000.00			
+	-----	+	-----	+	-----	+
----	+	-----	+	-----	+	

5 rows in set (0.00 sec)

mysql> notee